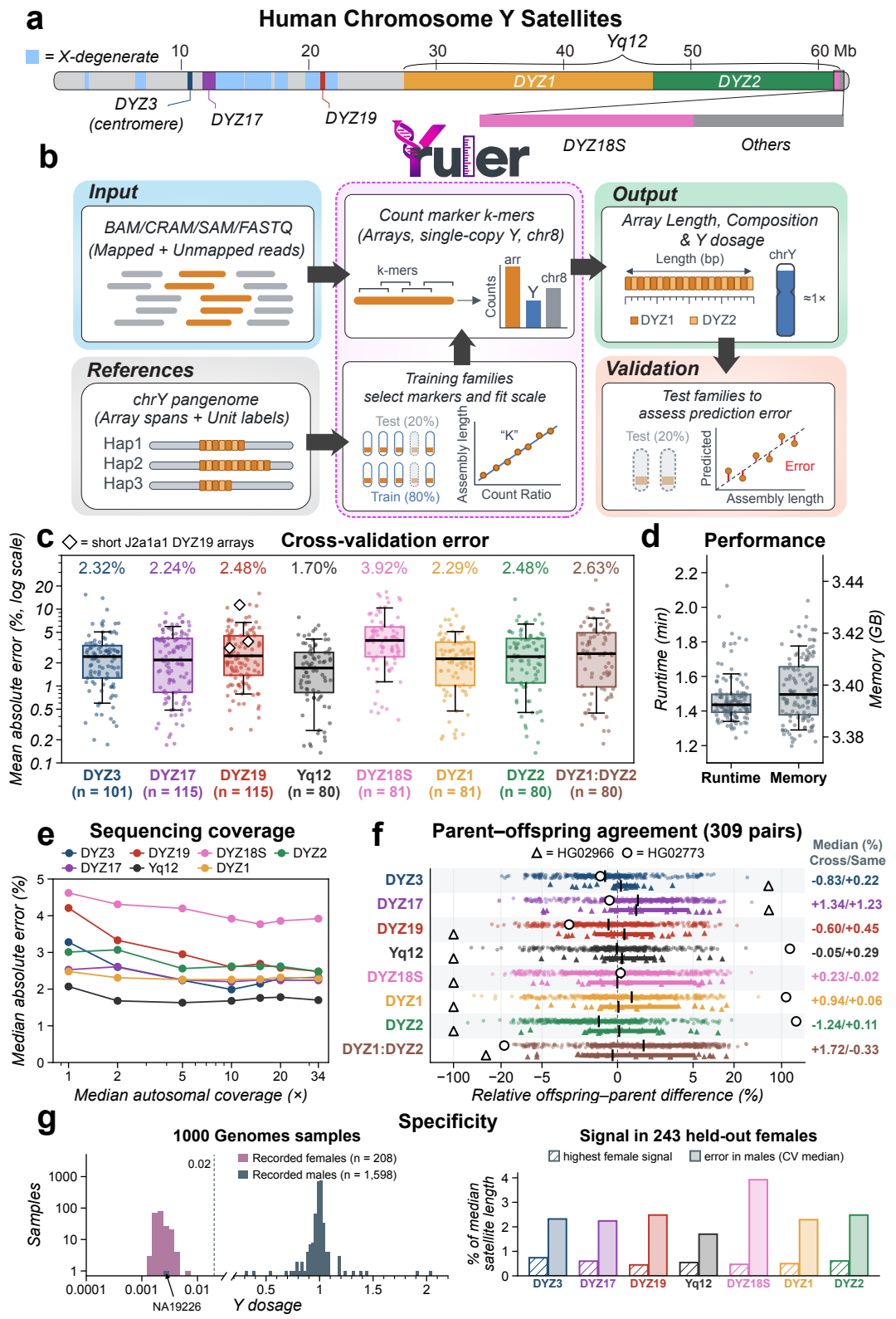

a
Human Chromosome Y Satellites
= X-degenerate
10
20
30
40
Yq12
50
60 Mb
DYZ1
DYZ2
DYZ3
(centromere)
DYZ17
DYZ19
DYZ18S
Others
b
Yuler
Input
BAM/CRAM/SAM/FASTQ
(Mapped + Unmapped reads)
References
chrY pangenome
(Array spans + Unit labels)
Hap1
Hap2
Hap3
Count marker k-mers
(Arrays, single-copy Y, chr8)
k-mers
Counts
arr
Y
chr8
Training families
select markers and fit scale
Test (20%)
Train (80%)
Assembly length
"K"
Count Ratio
Output
Array Length, Composition
& Y dosage
Length (bp)
chrY
≈1×
DYZ1
DYZ2
Validation
Test families to
assess prediction error
Test (20%)
Predicted
Error
Assembly length
c
◇ = short J2a1a1 DYZ19 arrays
Cross-validation error
Mean absolute error (%, log scale)
2.32%
2.24%
2.48%
1.70%
3.92%
2.29%
2.48%
2.63%
DYZ3 (n = 101)
DYZ17 (n = 115)
DYZ19 (n = 115)
Yq12 (n = 80)
DYZ18S (n = 81)
DYZ1 (n = 81)
DYZ2 (n = 80)
DYZ1:DYZ2 (n = 80)
d
Performance
Runtime (min)
Memory (GB)
Runtime
Memory
e
Sequencing coverage
DYZ3
DYZ19
DYZ18S
DYZ2
DYZ17
Yq12
DYZ1
Median absolute error (%)
Median autosomal coverage (×)
f
Parent–offspring agreement (309 pairs)
△ = HG02966 ○ = HG02773
Median (%)
Cross/Same
DYZ3 -0.83/+0.22
DYZ17 +1.34/+1.23
DYZ19 -0.60/+0.45
Yq12 -0.05/+0.29
DYZ18S +0.23/-0.02
DYZ1 +0.94/+0.06
DYZ2 -1.24/+0.11
DYZ1:DYZ2 +1.72/-0.33
Relative offspring–parent difference (%)
g
Specificity
1000 Genomes samples
Recorded females (n = 208)
Recorded males (n = 1,598)
Samples
NA19226
Y dosage
Signal in 243 held-out females
highest female signal
error in males (CV median)
% of median satellite length
DYZ3
DYZ17
DYZ19
Yq12
DYZ18S
DYZ1
DYZ2